# US Automobile Demand Estimation: Willingness to Pay & Price Elasticities using BLP Logit-IV

Berry, Levinsohn & Pakes (1995) US car dataset, 1971-1990

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## Part 4: Results and takeaways

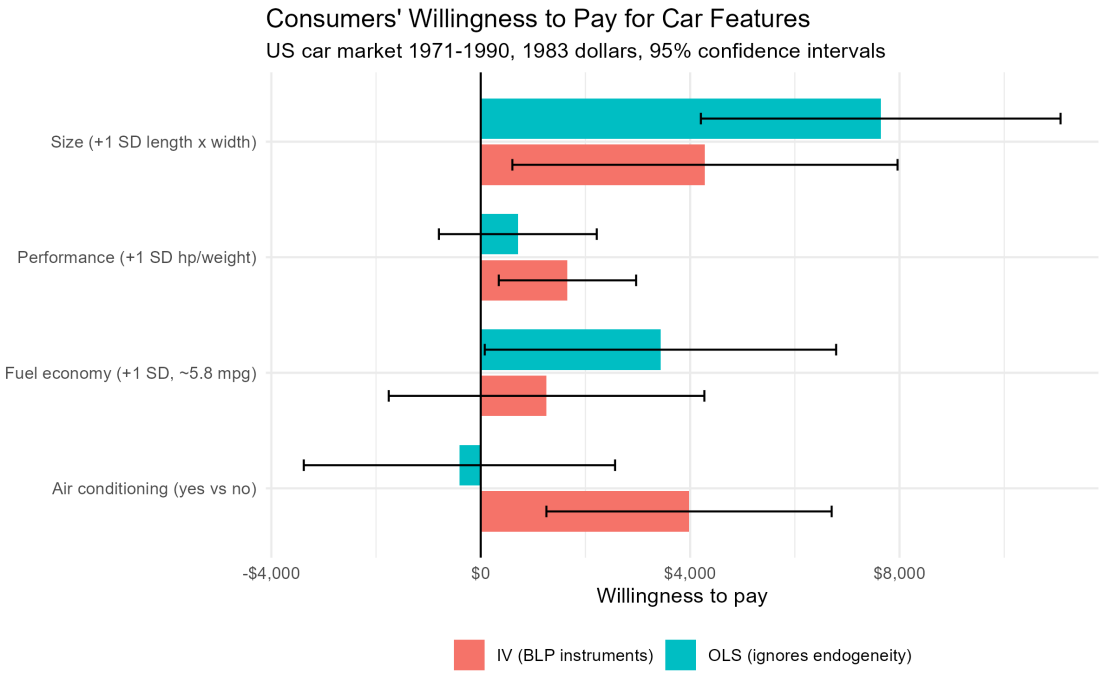



*Figure: Willingness to pay for car features (1983 $), OLS vs IV with 95% confidence intervals.*

| Feature | OLS | IV (BLP instruments) | 95% confidence interval (IV) |
|---|---|---|---|
| Air conditioning (yes vs no) | -$407 | $3,979 | $1,260 to $6,700 |
| Size (+1 SD) | $7,641 | $4,283 | $600 to $7,960 |
| Performance, hp/weight (+1 SD) | $709 | $1,657 | $350 to $2,970 |
| Fuel economy (+1 SD, ~5.8 mpg) | $3,433 | $1,257 | -$1,750 to $4,260 |

*Table: Willingness to pay for specs, OLS vs IV (1983 $). +1 SD = one standard deviation increase, i.e. a typical step up in that spec (e.g. +5.8 mpg).*

| Model | Price coefficient | Avg. own-price elasticity | Cars with inelastic demand |
|---|---|---|---|
| OLS (year FE) | -0.089 | -1.05 | 67.4% |
| IV (year FE) | -0.155 | -1.82 | 21.2% |

*Table: Price sensitivity, OLS vs IV.*

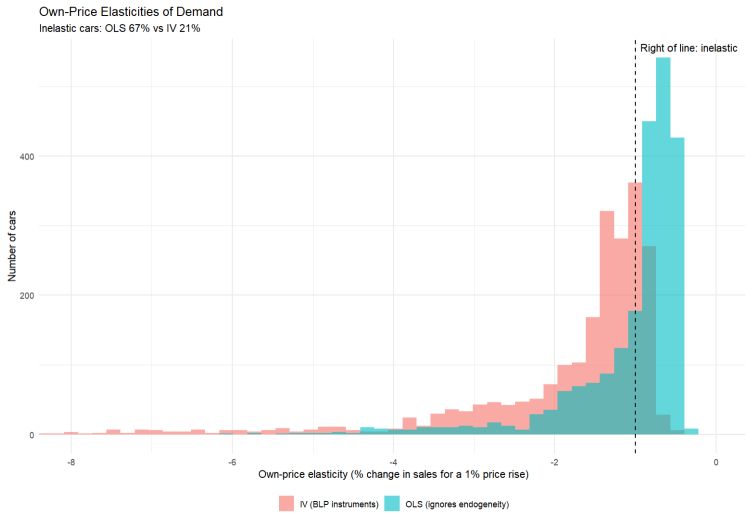



*Figure: Own-price elasticities, OLS vs IV.*

- OLS underestimates price sensitivity since better cars (unobserved quality) are both pricier and sell more. After instrumenting price, consumers come out ~75% more price sensitive and the share of cars with inelastic demand (not possible for a profit maximising firm) drops from 67% to 21%.
- OLS even gives a negative WTP for AC (-$407), IV corrects it to ~$4,000.
- Size and AC are the most valued specs, WTP for fuel economy is not statistically significant.
- These estimates can be directly used for pricing a spec, deciding discounts (using elasticities) and also to back out marginal costs of rival firms from their prices.
- Limitation: simple logit assumes same taste for everyone, so elasticities rise mechanically with price and implied markups are almost flat (~$6,500 per car). Next step is the full BLP random coefficients model.
- Note: the estimates are based on 1971-1990 US car data (in 1983 $), so the actual numbers are unlikely to be relevant for today's market. The aim here is to show the method, which can be applied to any recent dataset.

*[Check it out for full report, R code and full dataset -> github repo]*

## Part 1: Data and EDA

- Used the classic BLP (1995) US car dataset from 'BLPestimatoR' package in R: 2217 car models, 20 yearly markets (1971-1990) and 26 manufacturers, no missing values.
- Variables used: price ($1000s, 1983 $), hpwt (hp per 10 lbs of weight), air (AC dummy), mpg (in tens of miles per gallon), space (length x width) and market share.
- Each year ~89% of households don't buy a new car, which is taken as the outside option.

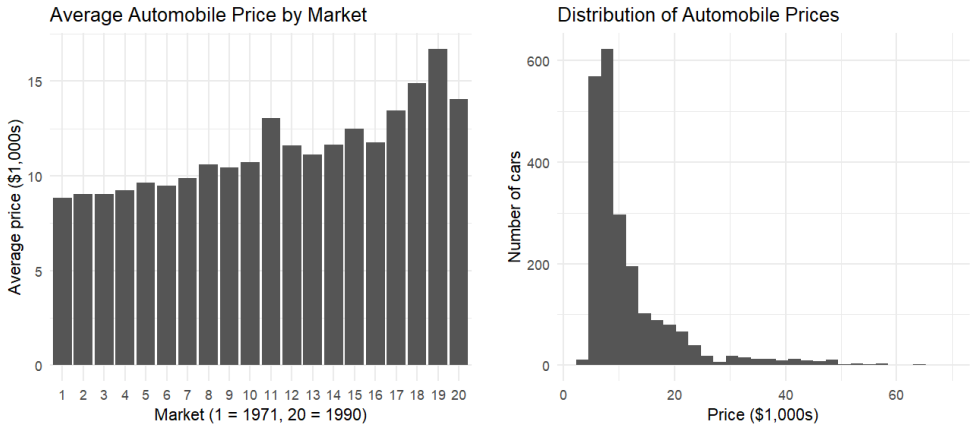



*Figure: Average car price by market (year), rising over 1971-1990 (left), and right-skewed distribution of car prices (right).*

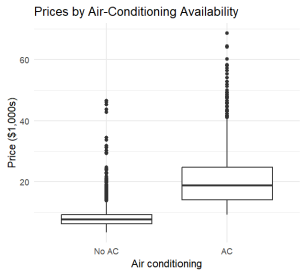



*Figure: Cars with AC are much pricier, part of which is unobserved quality (i.e. the endogeneity problem).*

## Part 2: Logit demand model

$$ln(s_j) - ln(s_0) = -\alpha\, p_j + \beta_1\, hpwt_j + \beta_2\, air_j + \beta_3\, mpg_j + \beta_4\, space_j + \xi_j$$

- Generated the outside good share $s_0$ = 1 − sum of shares for each year, and the dependent variable $\ln(s_j) - \ln(s_0)$ (Berry, 1994).
- Ran OLS with year fixed effects (to absorb macro shocks common to all cars in a year), standard errors clustered by manufacturer.
- WTP for a spec = $\beta_k / \alpha$ and own-price elasticity = $-\alpha\, p_j\, (1 - s_j)$.

## Part 3: Instrumental variables (BLP instruments)

- Price is endogenous since ξ (unobserved quality) affects both the price and the market share.
- Created BLP instruments, i.e. number of other cars of the same firm and the sum of their specs in the same year. Rival firms' sums were dropped since with year FE they are collinear with the

own-firm sums.

- Ran 2SLS with year FE. First stage F = 69.4 (strong instruments), Wu-Hausman p < 0.001 (price is endogenous), Sargan rejects (likely due to the simple logit).